DEVELOPMENT OF A 3-COMPONENT SEISMOMETER BASED ON AN MMA7361L MEMS ACCELEROMETER WITH SINGLE-SUPPLY CONFIGURATION FOR LOW FREQUENCY SEISMIC WAVE ACQUISITION

ABSTRACT

Seismic data acquisition is an important method for monitoring volcanic activity as well as for oil and gas exploration. This process generally uses instruments such as seismometers or velocity geophones as sensors to detect seismic waves; however, these devices are relatively expensive. This study aims to develop an alternative seismometer based on a MEMS (Micro-Electro-Mechanical System) accelerometer sensor of type MMA7361L. The sensor output, in the form of acceleration, is integrated to obtain the velocity of seismic waves. The system is designed using an MCP6002 operational amplifier configured as an integrator, voltage amplifier, filter, and buffer. The test results show that the system is capable of measuring seismic wave velocity in three components (x, y, and z axes) with an optimal frequency range of 1 Hz to 10 Hz and an overall frequency range of 0.1 Hz to 100 Hz. However, the noise in the MMA7361L output is still relatively high, which affects the measurement accuracy in the frequency range of 0.1 Hz to 4 Hz on the x and y axes. Therefore, the developed system has significant potential as a low-cost alternative seismometer, although further improvements are required for noise reduction.

Keywords : Seismometer, MEMS accelerometer, MMA7361L, seismic data acquisition, signal integration, low-cost sensor.